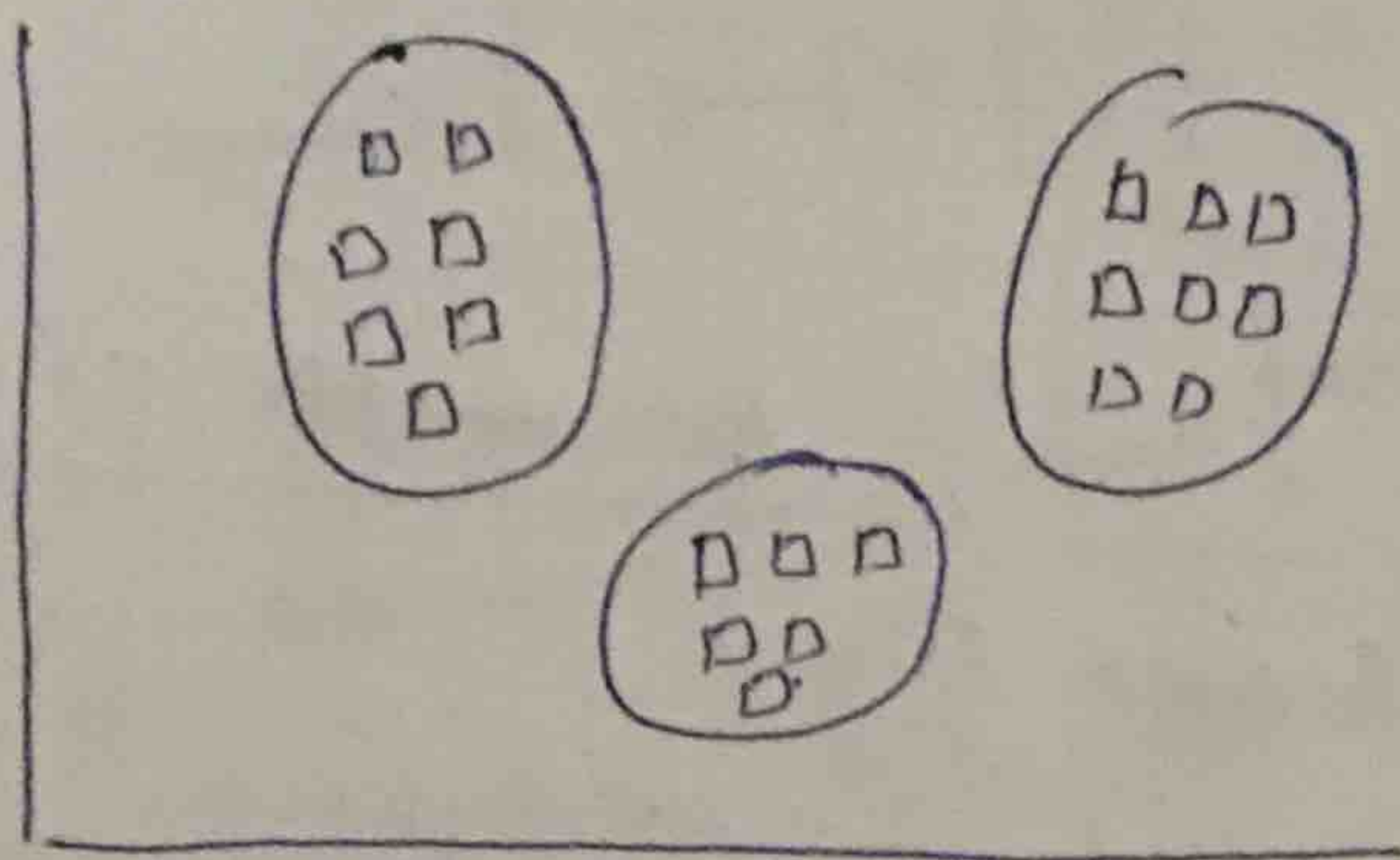


Clustering

Cluster Analysis or clustering is the task of assigning a set of objects into groups (called clusters). Objects in clusters are similar to each other. They are dissimilar to objects outside the clusters.



○ ○ ○ → outlier.

Clustering is also called ~~as~~ unsupervised learning because unlike classification clustering does not rely on predefined classes and class labels. We do not know the class labels and may not know the number of classes.

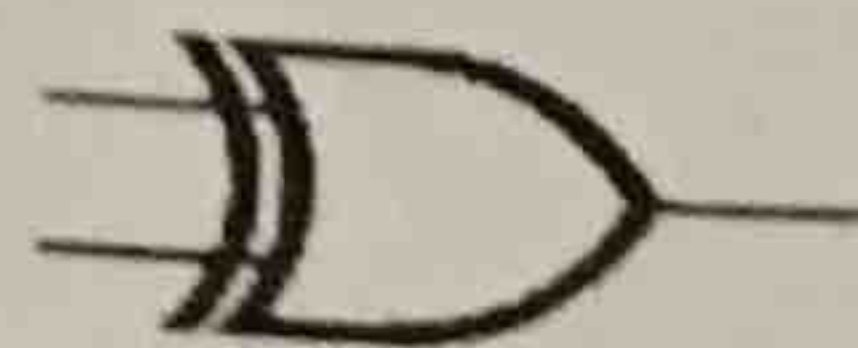
→ The distance between the points in clusters is less than the distance between a point in the cluster and any point outside it.

In given diagram, two or more objects belong to same cluster if they are close according to given distance. Outliers are objects that do not belong to any cluster or form cluster.

Library:- we can keep the book with similarity in styles & group them with several names.

(F): realization of Ex-OR gate

Symbol



Truth Table

2 Input EXOR gate		
A	B	$A \oplus B$
0	0	0
0	1	1
1	0	1
1	1	0

Logical expression:

$$Y = A \oplus B$$

Example of search clustering

(1) Search Engine

(2) Social N/W Analysis

used to recognise community within large group of people

(3) Marketing →

use clustering technique to cluster high salary customers and to target high salary clusters.

VHDL Code :

```
library ieee;
use ieee.std_logic_1164.all;

entity xorgate is
port(A,B:in std_logic;Y:out std_logic);
end xorgate;

shivam tecture shivam of nandgate is
begin
Y<=A xor B;
end shivam;
```

web search engine often return thousand of pages in response to broad query making it difficult for user to browse or identify relevant info.

clustering automatically group retrieved document in list of meaningful categories.

Example divide result of search for cell into groups like biology, botany, physics, fashion

Application of clustering In many fields the clustering can be used. 4.2

- 1) In an Economics application, it might be interested in finding countries whose economies are similar
- 2) In financial applications, we wish to find clusters of companies that have similar financial performance.
- 3) In marketing application, it might wish to ~~buy~~ find clusters of ~~companies~~ customers with similar buying behaviour.
- 4) In Medical application, we wish to find clusters of similar symptoms.
- 5) In crime Analysis, cluster analysis can be used to identify areas, where there are greater incidence of particular type of crime.
- 6) In Education, cluster analysis are used to identify groups of institution or students with similar properties.

→ Market Research

→ Data Analysis

→ Geography → spatial data analysis → identification of area of similar lands

→ outliers detection: The detection of credit card fraud & monitoring of criminal activities

Image processing

Requirements for cluster Analysis

4.3.

- (1) Scalability :- Large database contain million of objects.
Many clustering algorithm work well in small data objects.
- (2) Ability to Deal with different type of attribute :-
Application may require clustering such as binary, nominal (categorical) and ordinal data.
- (3) Ability to Deal with noisy Data :- Real world data sets contain outliers or missing, unknown or erroneous data. We need clustering method that are robust to noise.
- (4) Discovery of clusters with arbitrary shape :- Many clustering Algorithm determine cluster based on Euclidean or Manhattan distance measure. It is important to develop algorithm which can detect cluster of arbitrary shape.
- (5) Requirement for domain knowledge to determine input parameters.
Many clustering Algorithm require user to provide domain knowledge in the form of input parameters such as the desired number of clusters.
- (6) Interoperability And Usability :- Users want clustering results to be ~~interoperable~~ interpretable & Usable.

Measures of Similarity

Since clustering is the grouping of similar objects, some sort of measure that can determine whether two objects are similar or dissimilar. Two main types of measures are used to estimate this relation.

Distance Measure
Similarity Measure.

Many clustering methods use distance measures to determine similarity or dissimilarity between any pair of objects.

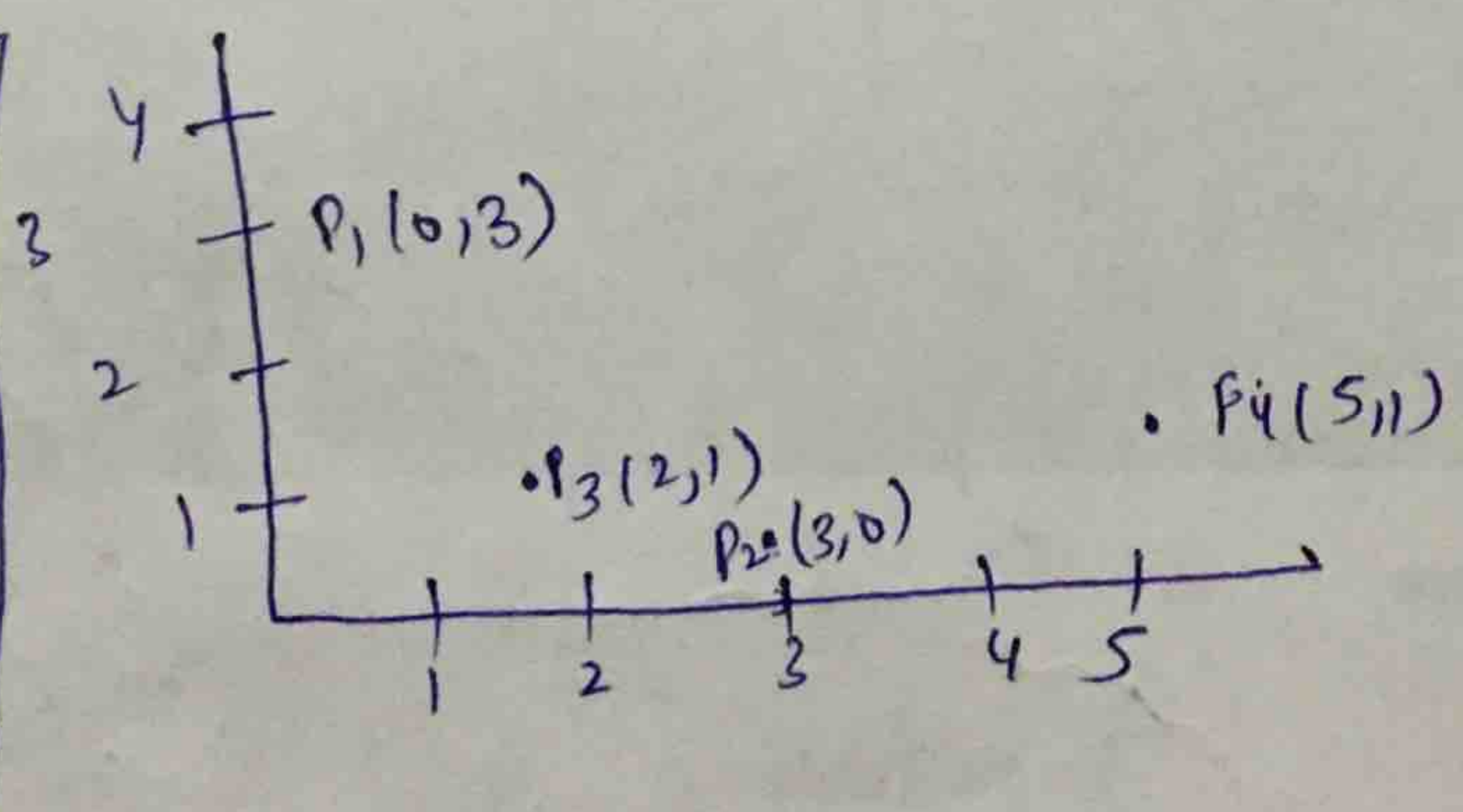
The most popular distance measures used are

- (1) Euclidean Distance
- (2) Manhattan Distance

Euclidean distance. Euclidean shortest distance corresponds to length of path b/w two samples.

$$d(i,j) = \sqrt{(x_{i1} - x_{j1})^2 + (x_{i2} - x_{j2})^2 + \dots + (x_{in} - x_{jn})^2}$$

Square roots of sum of squares of difference b/w coordinates.



	x	y
P1	0	3
P2	3	0
P3	2	1
P4	5	1

Euclidean distance b/w point P1 + P2.

$$= \sqrt{(0-3)^2 + (3-0)^2} = \sqrt{9+9} = \sqrt{18} = 4.2$$

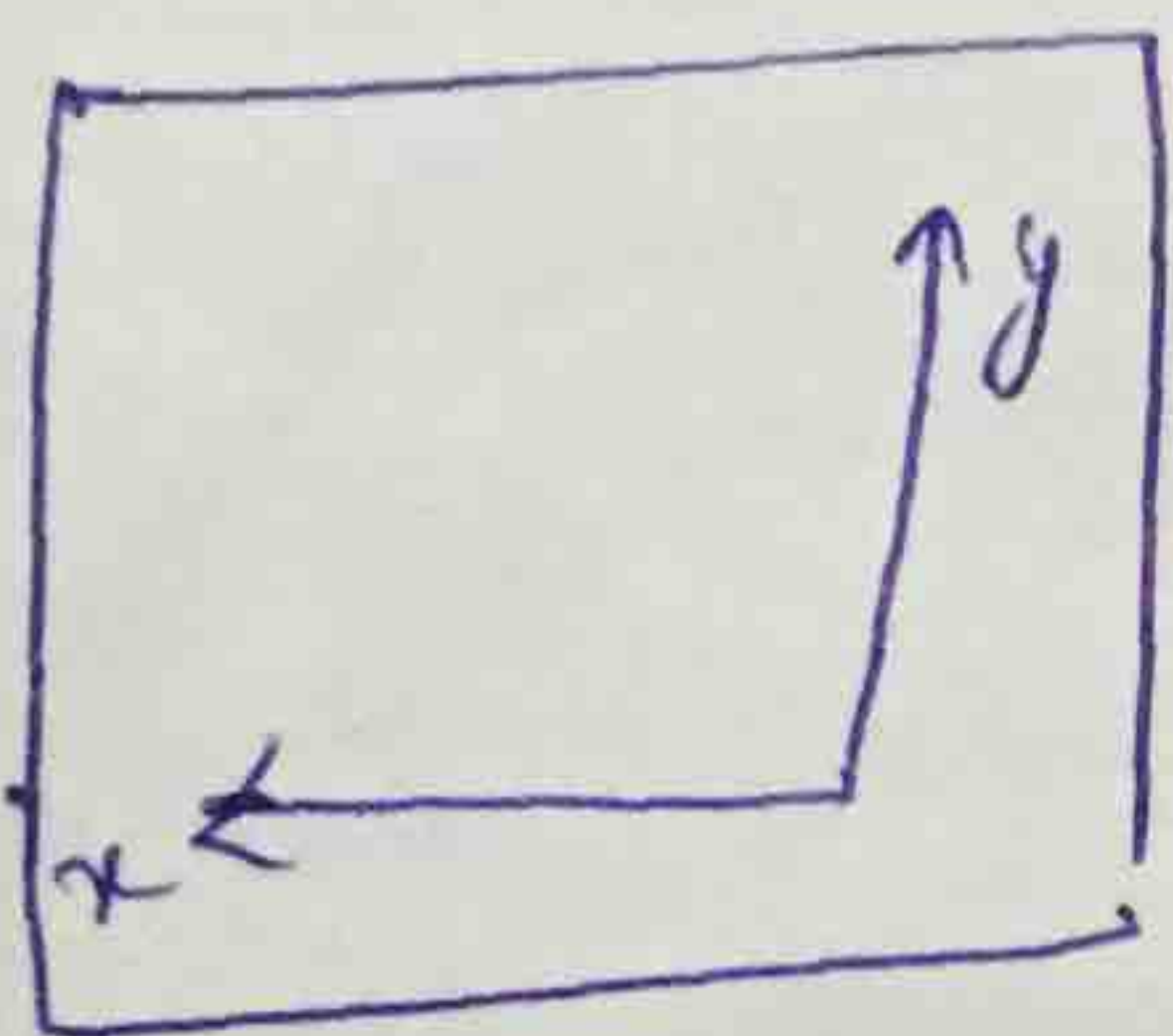
Manhattan distance

$$d(i,j) = |x_{i1} - x_{j1}| + |x_{i2} - x_{j2}| + \dots + |x_{in} - x_{jn}|$$

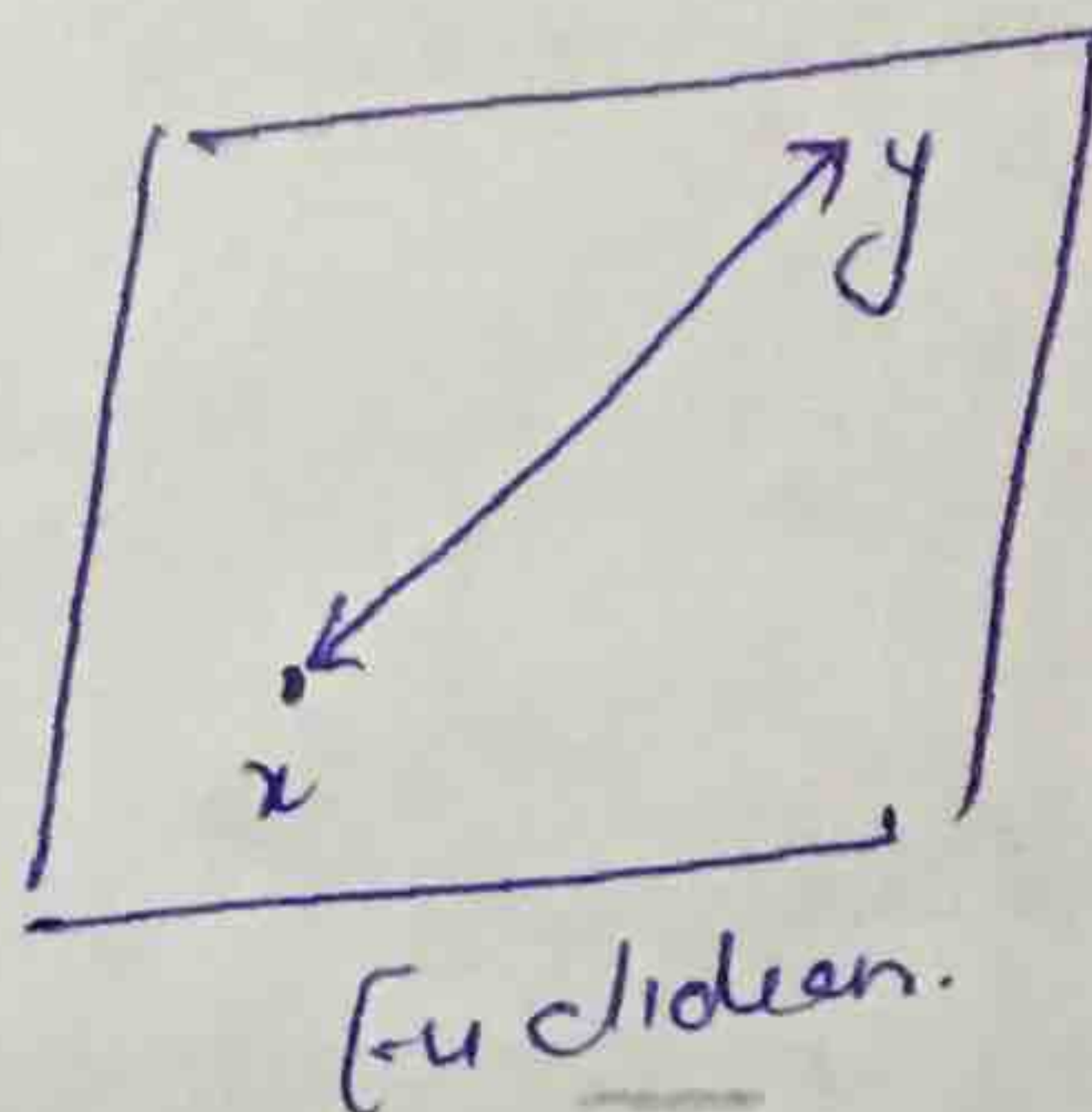
P_1 with coordinates (x_1, y_1) & Point P_2 with coordinates (x_2, y_2)

then $|x_1 - x_2| + |y_1 - y_2| \Rightarrow \sum_{i=1}^k |x_i - y_i|$

Manhattan distance is function compute the distance that would be travelled to get from one data point to other. The Manhattan distance b/w two items is the sum of differences of their corresponding points.



manhattan



Euclidean.

Euclidean distance: This distance examine the root of square difference between coordinates of a pair of objects.

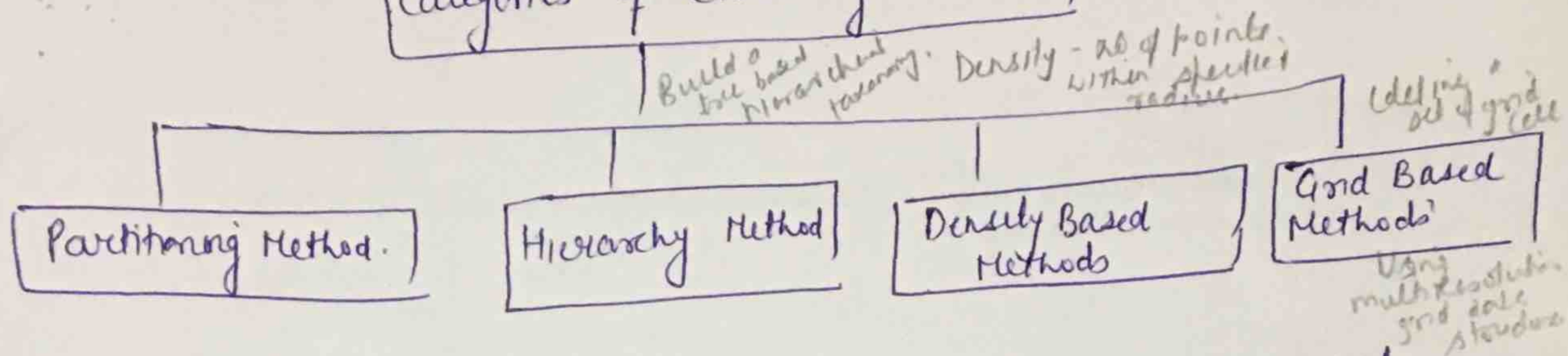
$$= \sqrt{\sum_{i=1}^k (x_i - y_i)^2}$$

	Cost	Time	Weight	Incentive
Object A	0	3	4	5
Object B	7	6	3	-1

$$d_{BA} = \sqrt{(0-7)^2 + (3-6)^2 + (4-3)^2 + (5+1)^2}$$

$$= \sqrt{49 + 9 + 1 + 36} = 9.747$$

Categories of Clustering Method.



Partitioning Algorithms :- Given a database D of n objects and partitioning method construct k partitions of the data where each partition represent a cluster and $k \leq n$.

The most well known & commonly used partitioning methods are k -Mean and k -medoids.

① k Mean :- A centroid Based Technique :- The k Mean algorithm takes the input parameter k and partition a set of n objects into k clusters so that resulting intraclustering similarity is high but the intercluster similarity is low. cluster similarity measured in regard to mean value of objects in cluster.

It is a type of square error algorithm. First it select randomly k of the objects each of which initially represent a cluster mean or centre. For each of the remaining objects, an object is assigned to the cluster to which it is most similar based on distance b/w object & cluster mean. It then compute new mean of each cluster.

$$E = \sum_{i=1}^k \sum_{p \in C_i} \text{dist}(p, c_i)^2$$

E is the sum of squared errors for all objects.
 p is the point in space representing given object.
 c_i is centroid of cluster C_i

Algorithm : K Mean

The k-Mean Algorithm for partitioning, where each cluster centre is represented by Mean value of object in cluster.

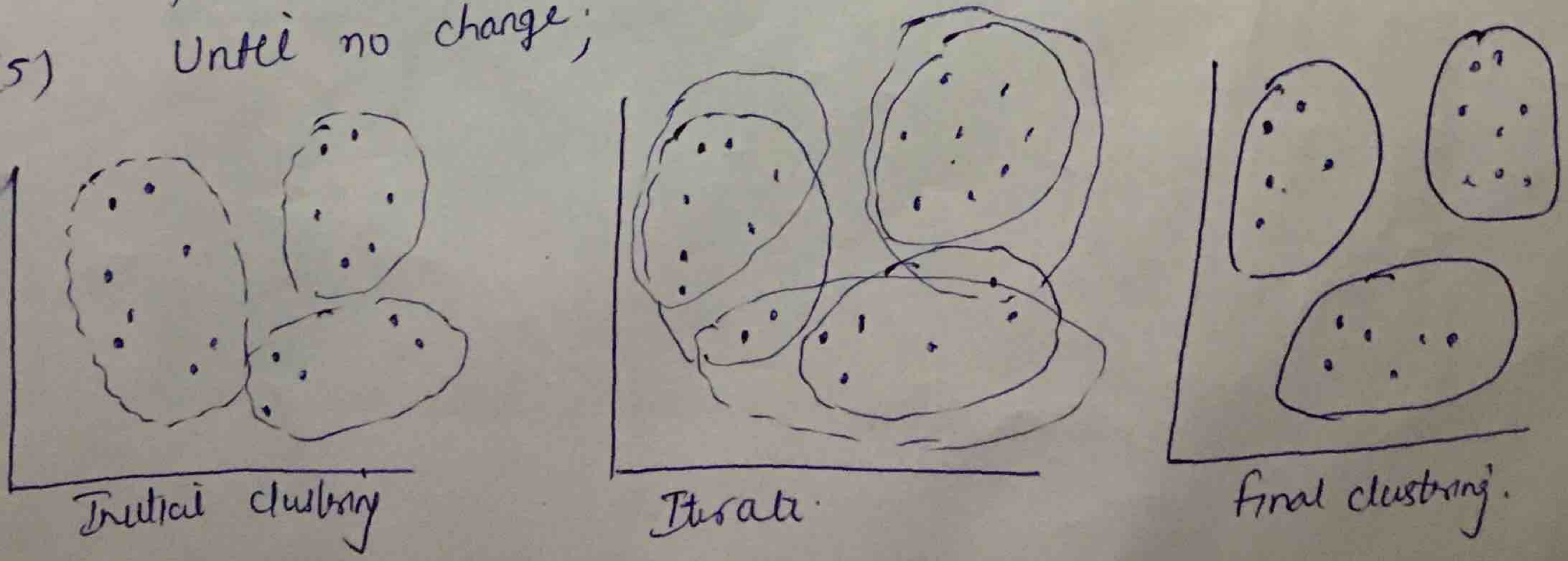
Input:

- k : the number of clusters
- D : a data set containing n objects.

Output: A set of k clusters.

Method:

- (1) arbitrary choose k objects from D as initial cluster centres.
- (2) Repeat
- (3) (or) assign each object to cluster to which object is most similar based upon mean value.
- (4) Update cluster mean i.e. calculate mean value of objects for each cluster;
- (5) Until no change;



4.8

Advantage: This Algorithm works well when the clusters are compact clouds that are rather well separated from one another.

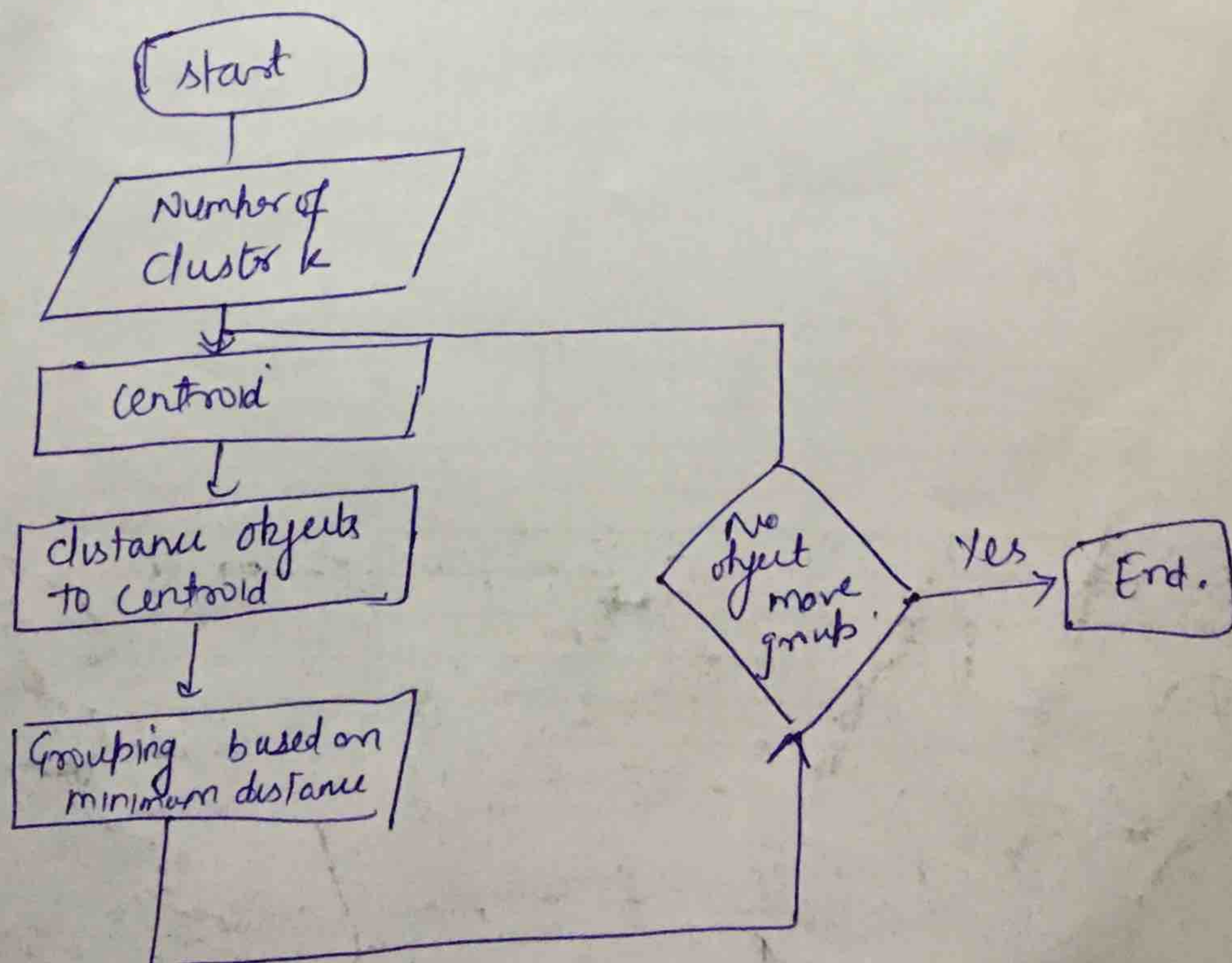
2) Complexity of algorithm is $O(nkt)$, n is total number of data points, k is number of clusters & t is the iterations.

Disadvantage: This algorithm is sensitive to noise & outliers data points because small numbers of such data influence mean value.

2) k Mean algorithm can be applied only when the mean of cluster is defined

3) The necessity for user to specify k in advance can be seen as disadvantage.

How Algorithm works:



Numericals on Subsetting

4.9

Suppose we want to group the visitors to a website using just their age as follow (one dimensional)

15, 15, 16, 19, 19, 20, 20, 21, 22, 28, 35, 40, 41, 42, 43, 44, 60, 61, 65.

Initial clusters

$$\text{Centroid } (C_1) = 16 \quad [16]$$

$$\text{Centroid } (C_2) = 22 \quad [22]$$

Iteration 1

$$C_1 = 15.33 \quad [15, 15, 16]$$

$$C_2 = 36.25 \quad [19, 19, 20, 20, 21, 22, 28, 35, 40, 41, 42, 43, 44, 60, 61, 65]$$

Iteration 2

$$C_1 = 18.56 \quad [15, 15, 16, 19, 19, 20, 20, 21, 22]$$

$$C_2 = 45.90 \quad [28, 35, 40, 41, 42, 43, 44, 60, 61, 65]$$

Iteration 3

$$C_1 = 19.50 \quad [15, 15, 16, 19, 19, 20, 20, 21, 22, 28]$$

$$C_2 = 47.89 \quad [35, 40, 41, 42, 43, 44, 60, 61, 65]$$

Iteration 4

$$C_1 = 19.50 \quad [15, 15, 16, 19, 19, 20, 20, 21, 22, 28]$$

$$C_2 = 47.89 \quad [35, 40, 41, 42, 43, 44, 60, 61, 65]$$

No change b/w Iteration 3 & 4. has noted.
By clustering two groups have been identified

15-28 & 35-65.

Algorithm is often run multiple times.

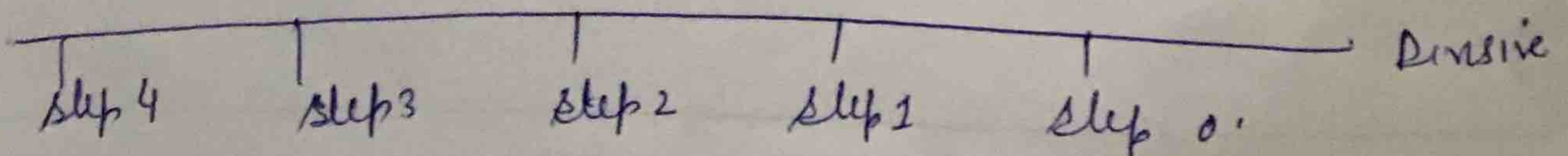
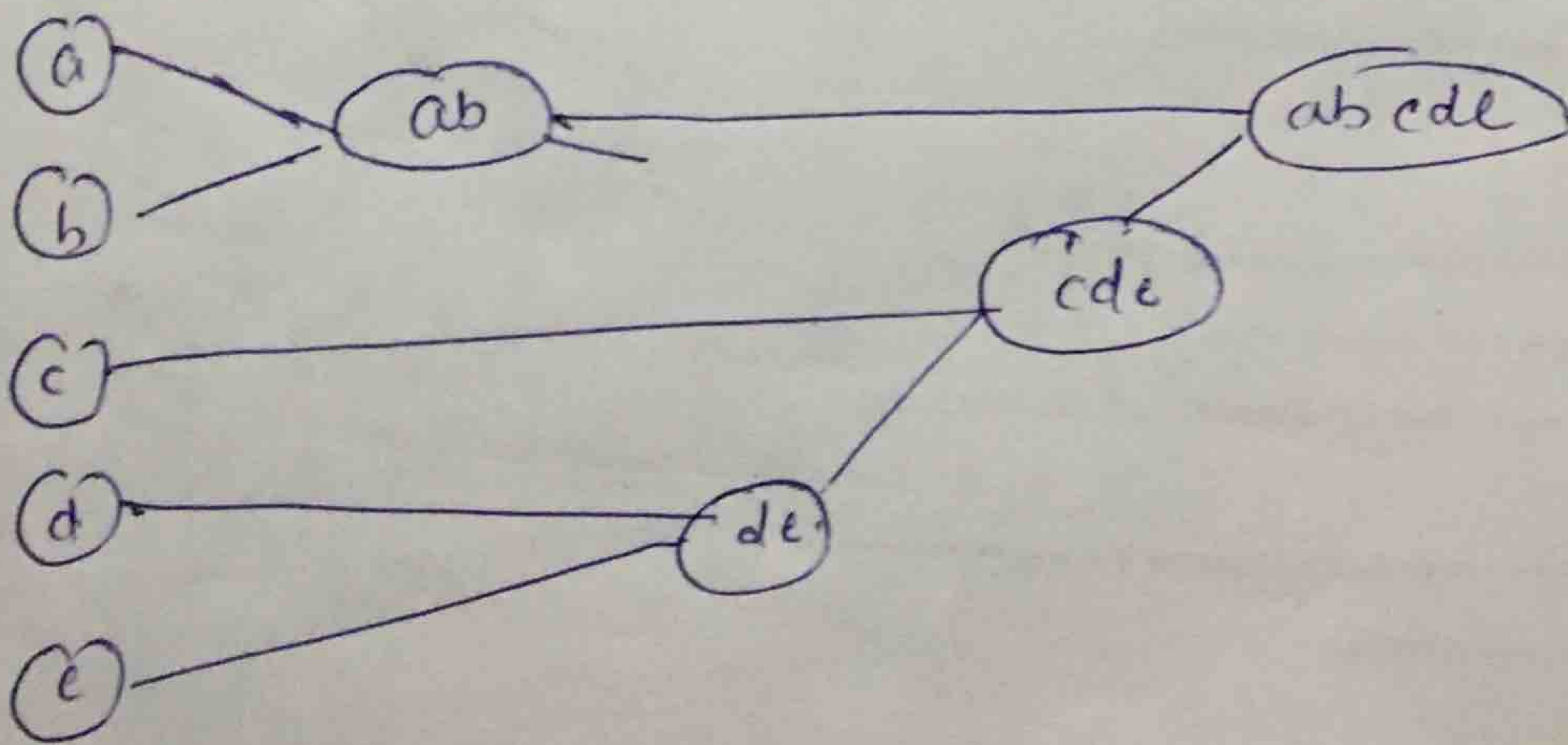
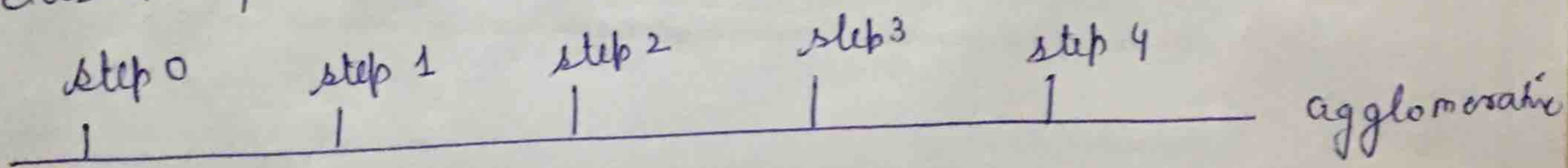
Hierarchical clustering

4.10

A Hierarchical clustering method work by grouping data object into tree of cluster. In hierarchical clustering the goal is to produce a hierarchical series of nested clusters. A diagram called dendrogram graphically represent this hierarchy and is an inverted tree that describe the order in which points are merged (bottom up view) or cluster are split (top down view). The basic approaches to generate hierarchical clustering

(i) Agglomerative - start with the points as individual cluster and at each step, merge the closest pair of cluster. (This bottom up approach).

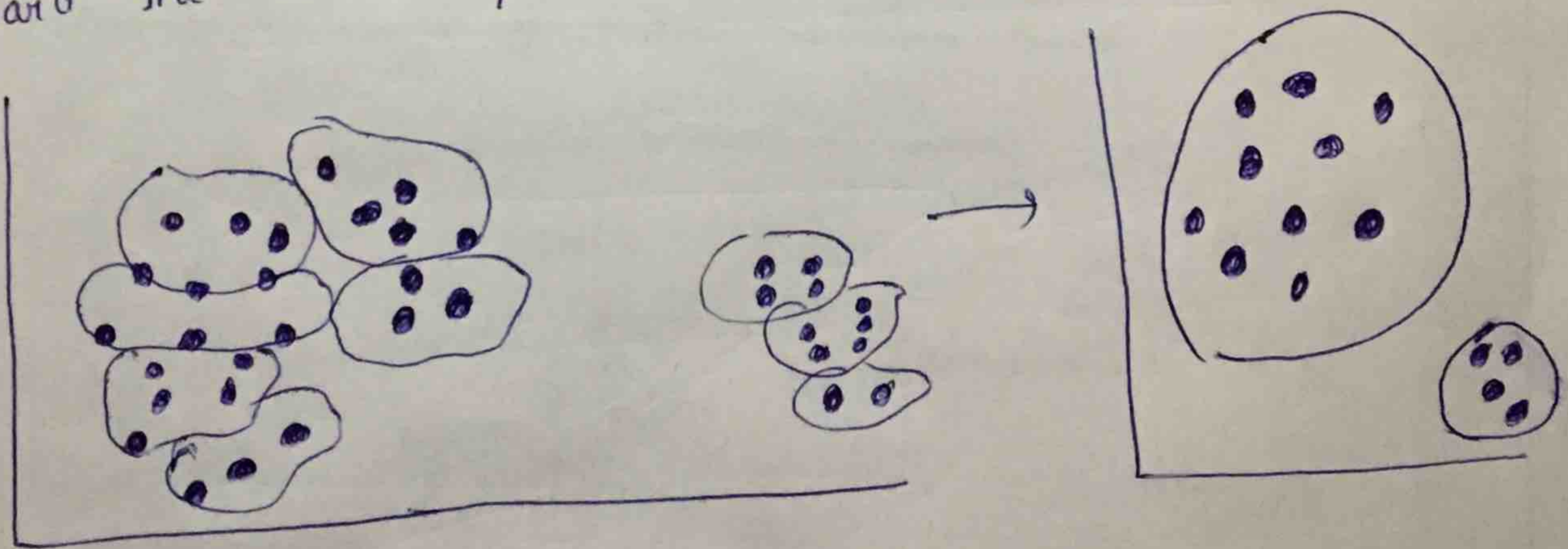
(ii) Divisive - start with one, all inclusive cluster and at each step, split a cluster until only singleton cluster of individual points remain.



CURE (Clustering using REpresentatives). 4.11
Most clustering Algorithm favour cluster with spherical shape and similar size. CURE is an alternative method. Which integrate hierarchical and partitioning algo. It overcome the Pbm of favouring cluster with spherical shape and similar size and is more robust with respect to outliers.

Instead of using a single centroid or object to represent a cluster, a fixed no of representative points in space are chosen. The representative points of cluster are generated by first selecting well scattered objects for clusters. At each step of Algo, the two cluster with close pair of representative point (Where each point in pair is from different cluster) are merged.

- Representative points are found by selecting constant no of points from cluster and then "shrinking" them toward the centre of cluster.



Density Based Method.

4,12

Density Based clustering locate region of high density that are separated from one another by region of low density. (representing noise). To discover cluster with arbitrary shape, density based clustering method have been developed.

Density = number of points within specified radius. (Eps).

Major features are.

- (i) discover cluster of arbitrary shape. (unknown shape).
- (ii) Handle noise
- (iii) one scan.

Two parameter are used.

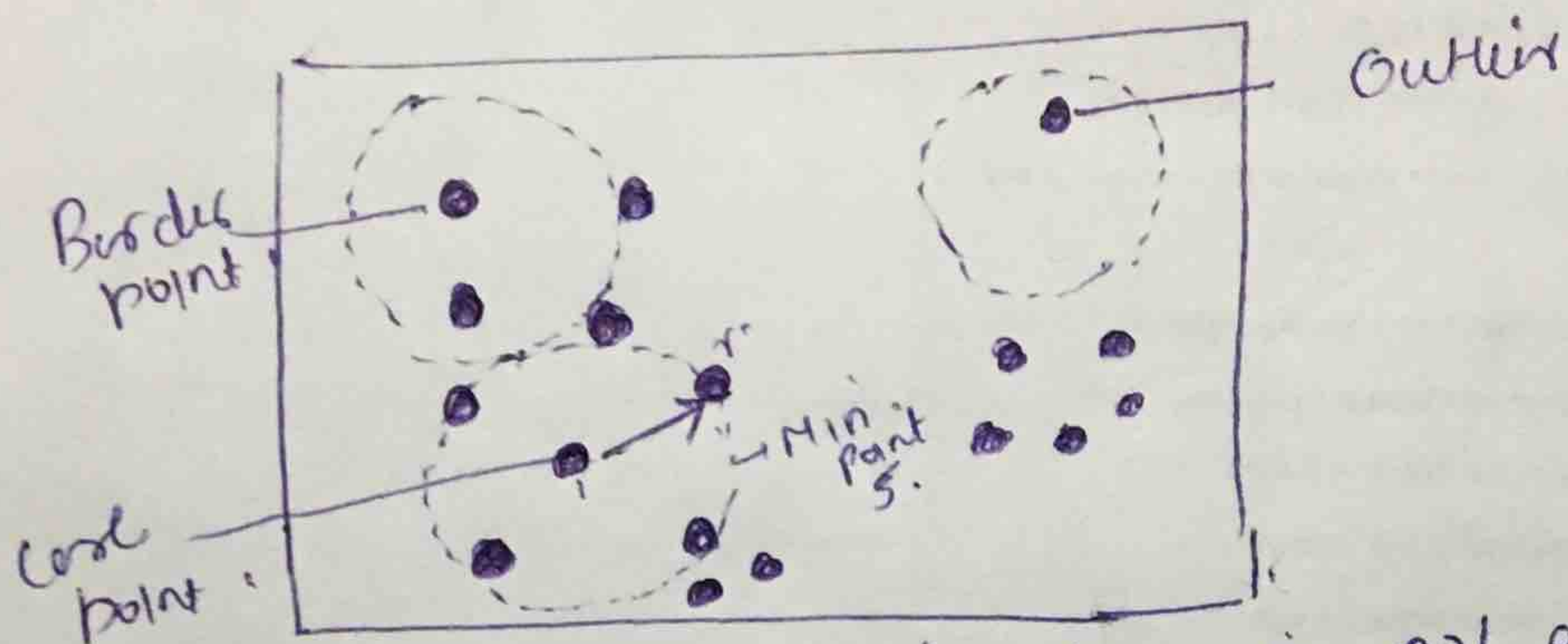
- (i) Eps : Maximum radius of neighbourhood.
- (ii) Min Pts : Minimum no of points in Eps - neighbourhood of that point.

[Region with high density of points depicts existence of cluster where as region with low density of points indicate cluster of noise or cluster of outliers. This Algo is particularly suited to deal with large datasets with noise and is able to identify cluster with different size and shape.

[DBSCAN → Density Based Spatial clustering of Applications with Noise)

OPTICS → Ordering point to identify clustering structure.

Principle → The basic idea is to continue growing ^{4.13} the given cluster as long as the density in neighbourhood exceed some threshold i.e. for each data point within given cluster, the radius of given cluster has to contain at least min number of points



Noise point → It is any point that is not a core point or border point.

Core point :- A point is a core point if it has more than specified number of points (min pts) within ϵ s.

Border Point → Border point has fewer points than min pts.

DBSCAN Algo

- (i) select a point P
- (ii) Retrieve all points density reachable from P w.r.t radius & min points
- (iii) If P is core point, cluster is formed.
- (iv) If P is border point, no points are density reachable from P and DBSCAN visit next point of database
- (v) Continue the process until all of the points have been processed.

An Example

4.14.

